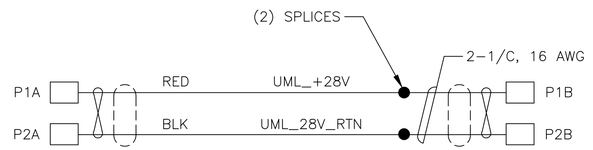
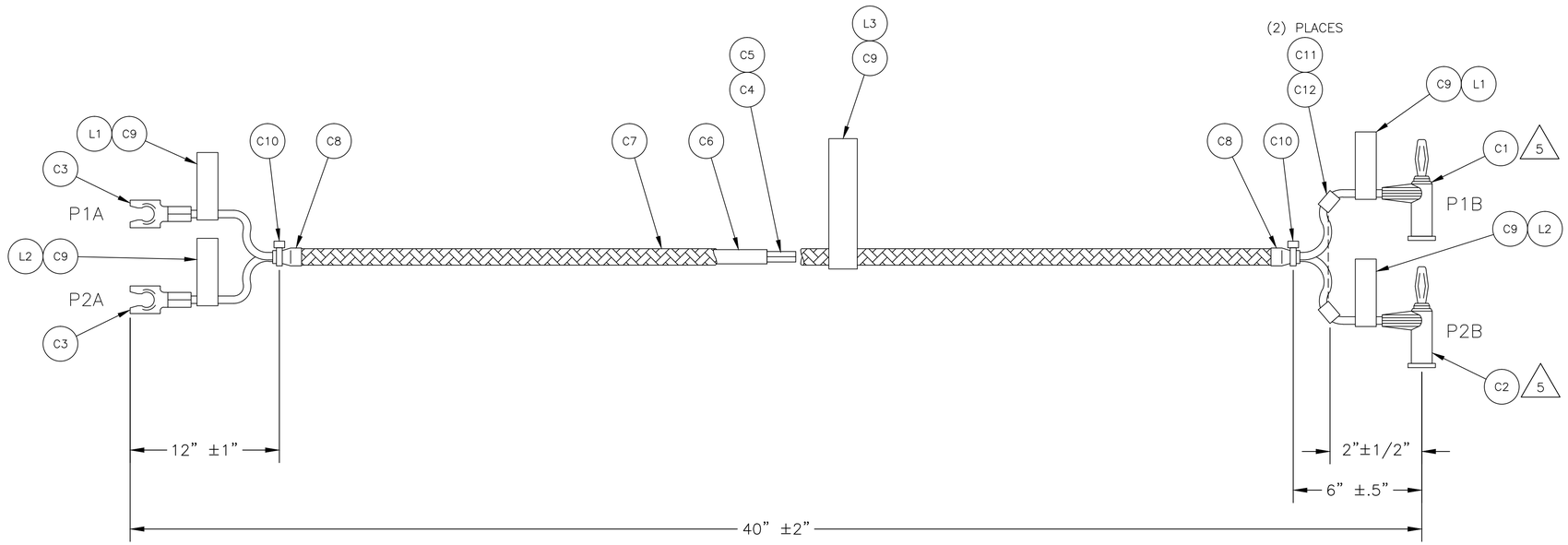
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	1	1825-2	PLUG BANANA, RED	POMONA
C2	1	1825-0	PLUG BANANA, BLACK	POMONA
C3	2	MU14-14FLK	FORK TERMINAL, 1/4" STUD, AWG 16-14	3M
C4	A/R	M22759/11-16-2	1/C, 16 AWG, PTFE INSULATED, RED	QPL
C5	A/R	M22759/11-16-0	1/C, 16 AWG, PTFE INSULATED, BLACK	QPL
C6	A/R	FMC-EXPHRZZZ-003-32	EXPANDABLE SLEEVING, 1/8" DIA, BLACK	BENTLEY HARRIS
C7	A/R	2140	TINNED COPPER BRAID, 3/16"	ALPHA
C8	A/R	THFE4-365012	HEAT SHRINK TUBING, 0.356" ID	SAINT-GOBAIN
C9	A/R	TZE-211	LABEL, P-TOUCH, 1/4", BLACK ON WHITE	BROTHER
C10	2	PLT1M-M	CABLE TIE, 4" NYLON	PANDUIT
C11	2	M14BCK	TERMINAL, BUTT SPLICE, 16-14 AWG	3M
C12	A/R	P-224X.5	TAPE, KAPTON, 0.5" WIDE	PERMACEL
C13	-	-	-DELETED-	-

LABEL LIST	
ITEM	INSCRIPTION
L1	UML +28V
L2	UML 28V RTN
L3	S1101EGA4569 SN xxxx



SCHEMATIC



NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND THE FOLLOWING NOTES:
   a. WIRE TWIST SHALL BE BETWEEN 8 AND 16 TIMES THE OUTER DIAMETER OF THE HARNESS UNLESS OTHERWISE NOTED. (FROM NASA-STD-8739.4 PARA 7.3.20)
5. ASSEMBLE BANANA PLUGS PER MANUFACTURER'S PROCEDURES.
6. TEST PER NASA-STD-8739.4 PARA 18.2.6 USING THE FOLLOWING:
   CONTINUITY: < 5 OHMS
   DWV: VOLTAGE: 1500 ±75VDC
   CURRENT: < 1 MILLIAMP
   DURATION: < 1 MINUTE
   IR: VOLTAGE: 500 ±75VDC
   RESISTANCE: < 100 MEG OHMS
   DURATION: < 1 MINUTE
7. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

CRIMP TOOL LIST				
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL	REF. DES.
MU14-14FLK	TR-490	N/A	N/A	P1A, P2A, P1B, P2B
M14BCK	TR-490	N/A	N/A	BUTT SPLICES

TENSILE PULL STRENGTH LIST			
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
3M	MU14-14FLK	16	50.0
3M	M14BCK	16	50.0

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-411.	MH 9/28/15 DS 9/28/15 TG 9/29/15

REVISION HISTORY

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 9/29/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 11/21/14	QUALITY	D. SEELEY 11/25/14
REVIEWER	J. BRUEWER 11/25/14	MATERIALS	M. WILKINSON 11/24/14
MECHANICAL	A. DRAKE 11/21/14	PROJ LEAD	B. BOROWSKI 11/25/14
ELECTRICAL	T. GOBELI 11/21/14		
STRUCTURAL	C. WERNER 11/21/14		
SAFETY	W. BORRELLI 11/21/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
UML SIMULATOR EMI
RADIATED BREAK OUT HARNESS

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4569	A

SCALE	NONE	INITIAL REL DATE	11/26/14	SHEET	1	OF	1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EGA4569_REV_A.DWG

AutoCAD 2012

DWG NO S1101EGA4569 SH 1 REV A